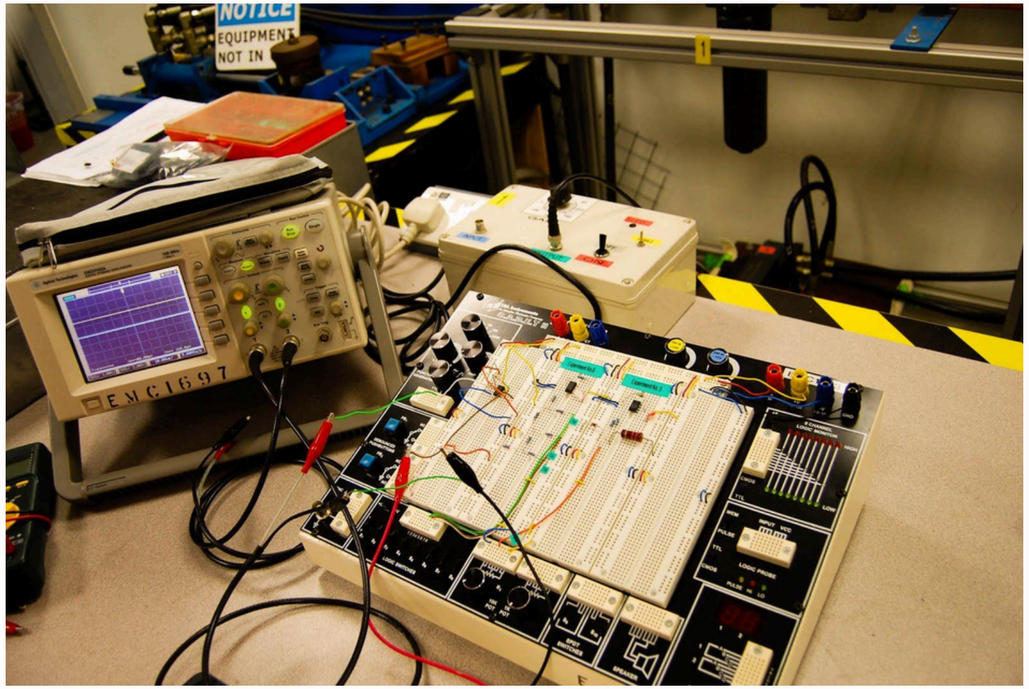
LAB BENCH DEBUG KIT

A Portable Circuit Testing, Protection & Troubleshooting System



WHAT IS MY PROJECT?

BURN GUARD

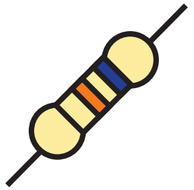


Problem: IC damage from overcurrent

Solution: Detects overcurrent and cuts off the power, with warning.

Working: LM 393 + Current Sensor → Overcurrent → Power OFF + Alert

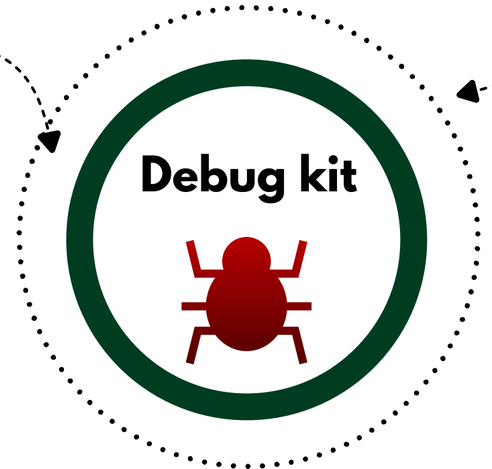
RESISTOR TOUCHPAD



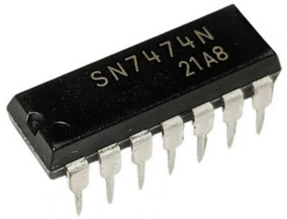
Problem: Unknown resistor value

Solution: Quickly identifies the approximate resistance range using voltage comparison and LEDs.

Working: Voltage Divider + LM 393 → Resistance Range → LED



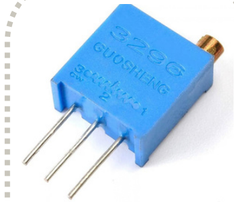
BURN GUARD



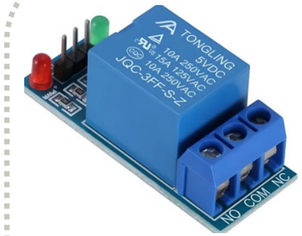
7474 — dual D flip-flop



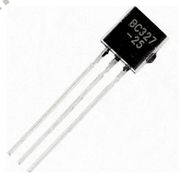
LM393-dual comparator



10K trimpot — variable resistor



5V relay module



BC327



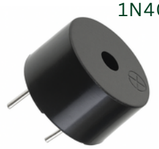
1N4007 diode



Push button



Red LED



5V active buzzer

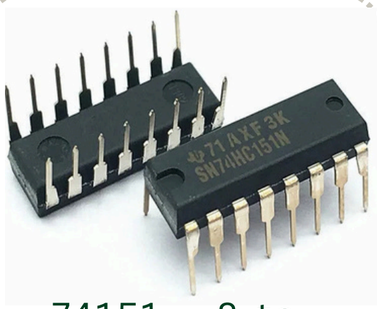


4.7 Ω 1W resistor

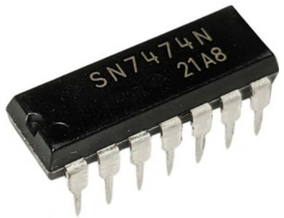


10 µF capacitor

IC TESTER



74151 — 8-to-1 multiplexer



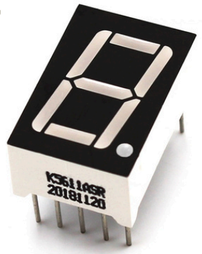
7474 — dual D flip-flop



7486 — quad XOR gate



7447 — BCD to 7-segment decoder



Common anode 7 segment display



14 pin ZIF socket



4 way DIP switch

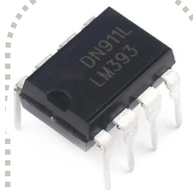
LOGIC PROBE



7400 — quad NAND gate



7474 — dual D flip-flop



LM393 — dual comparator



probe tip



LED



3.9 kΩ 1% resistor

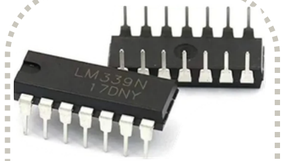


0.1 µF capacitor



Crocodile clip wire

RESISTOR TOUCHPAD



LM339 — quad comparator

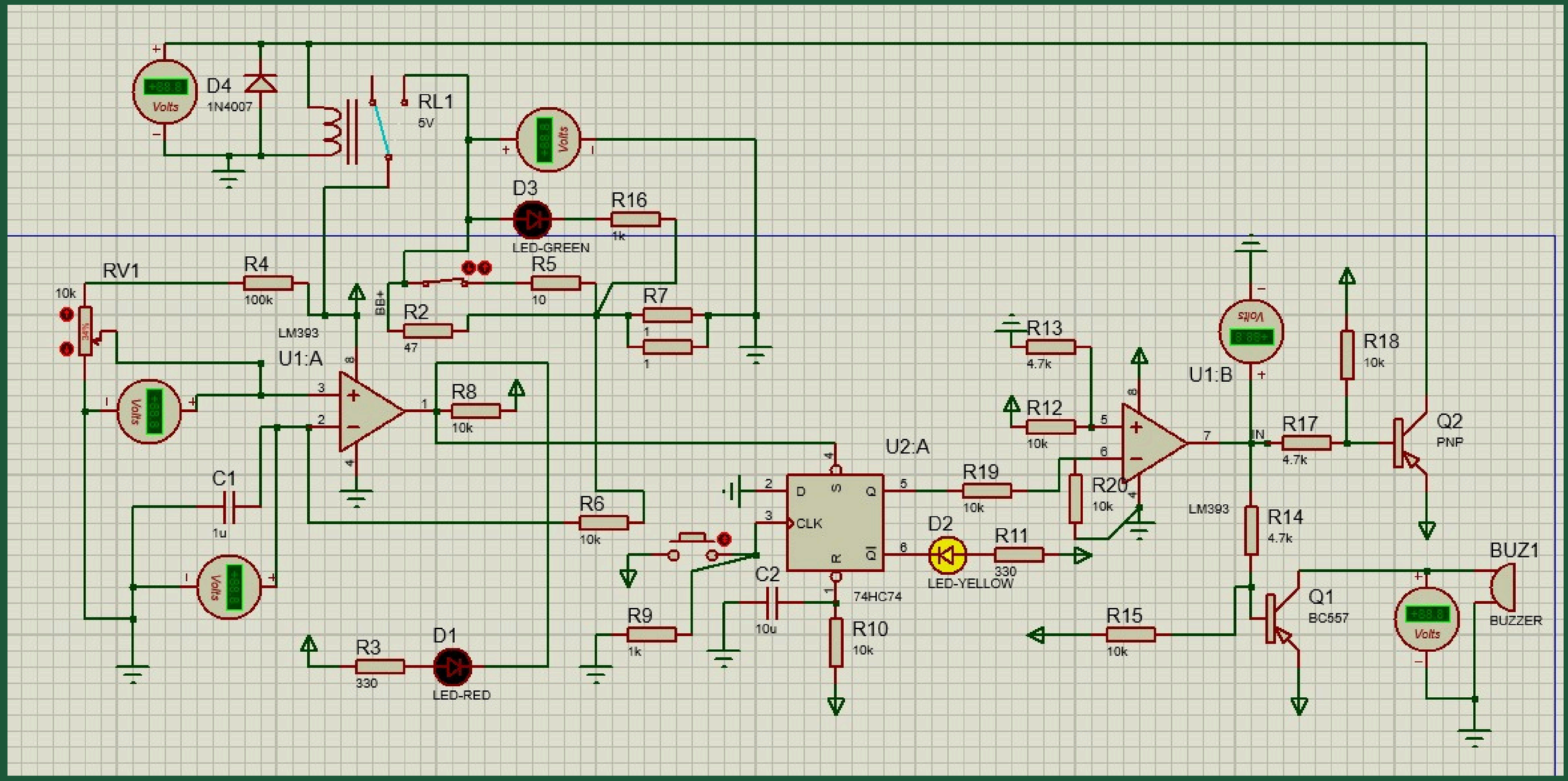


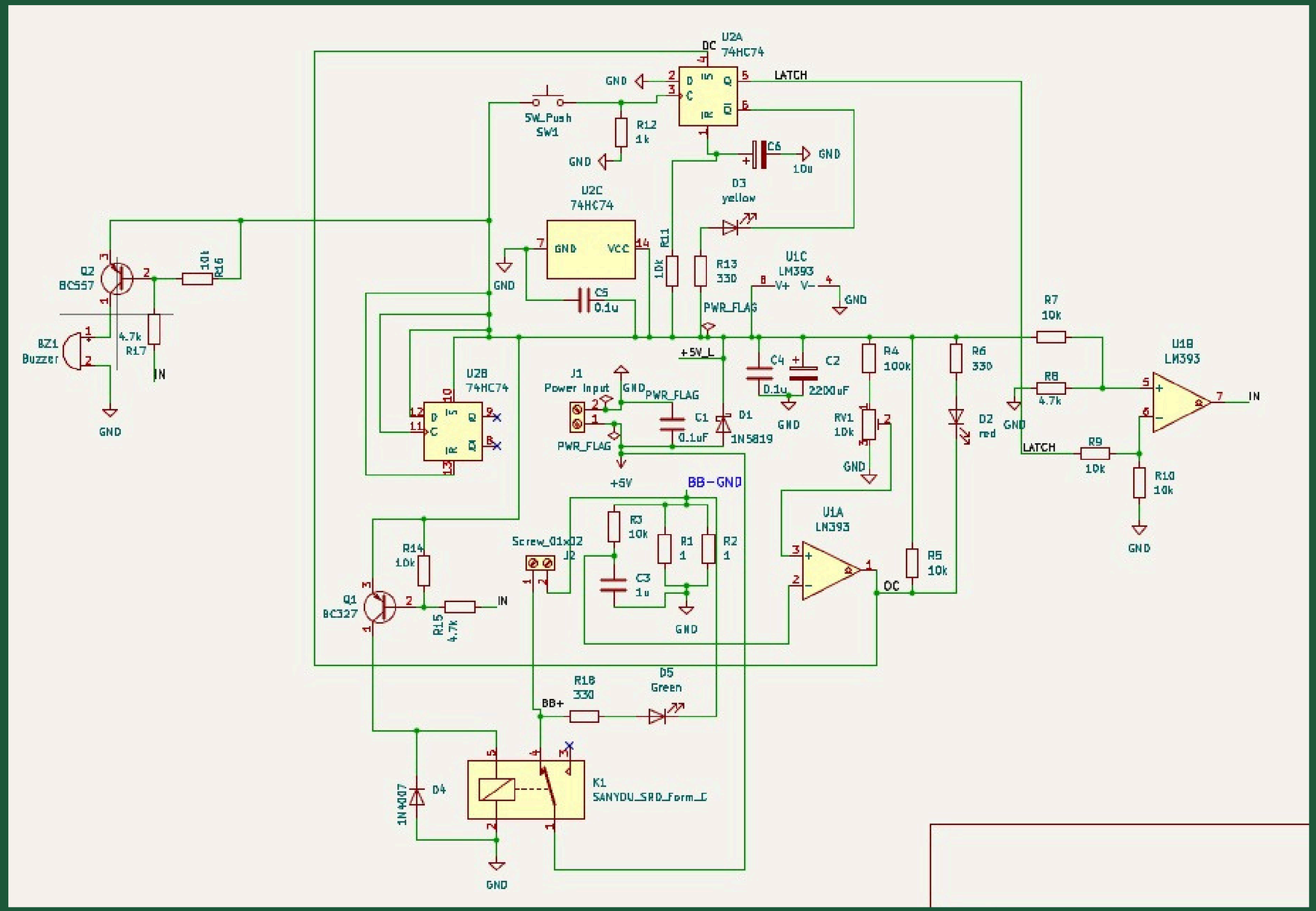
Copper pad



LED

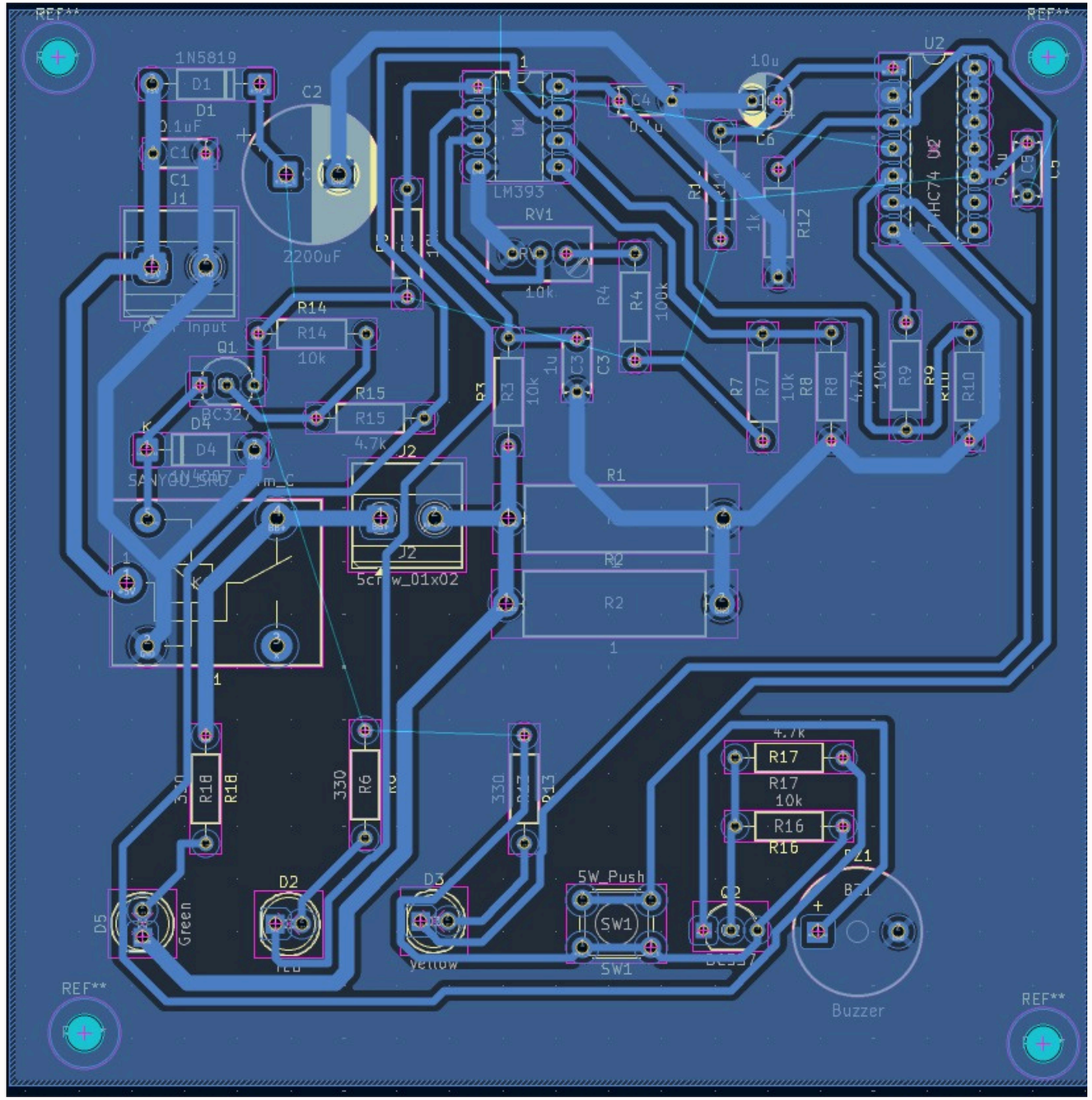
BURN GUARD SIMULATION





BURN GUARD

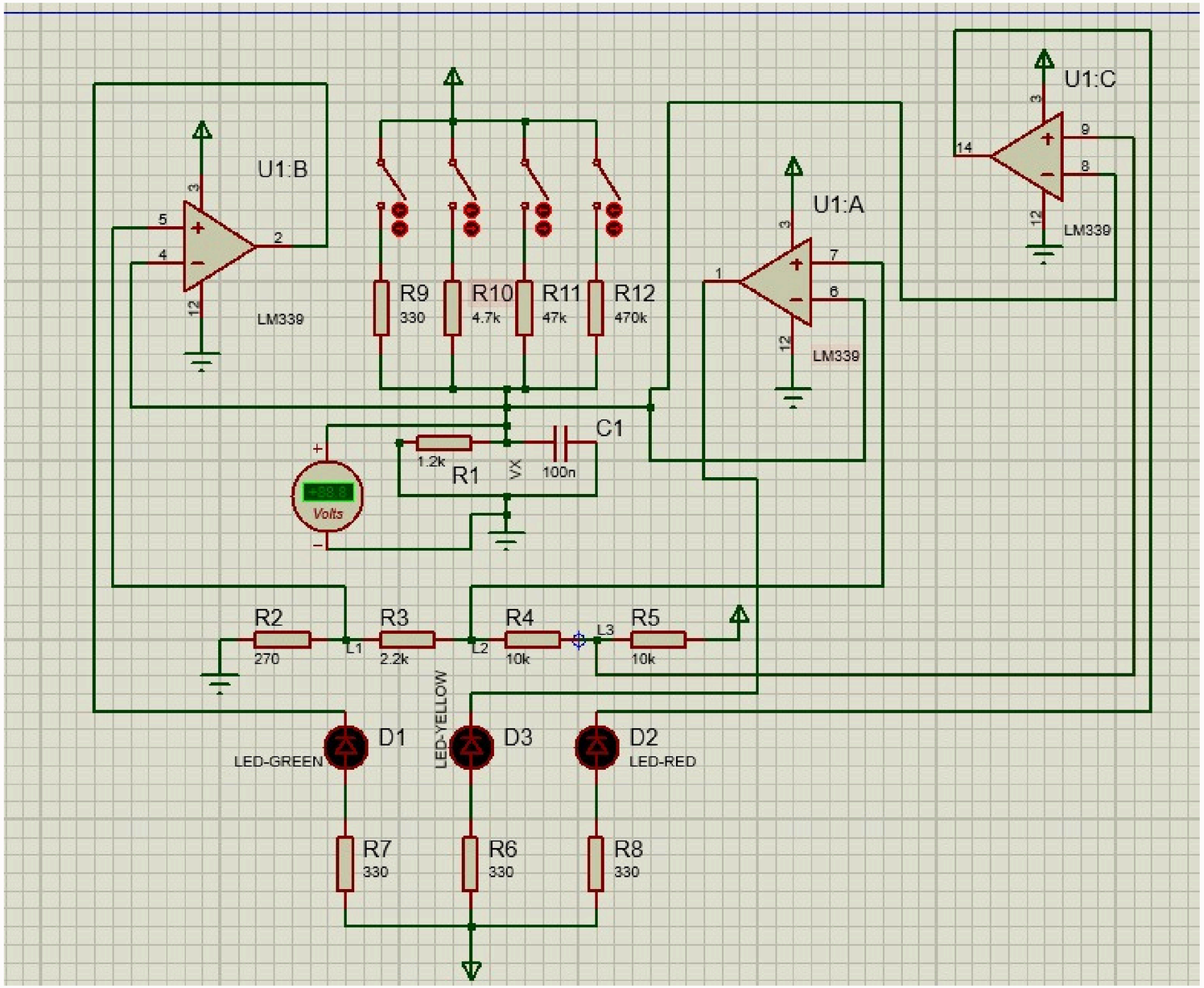
SCHEMATIC



- PCB BURN GUARD

RESISTOR TOUCHPAD

SIMULATION

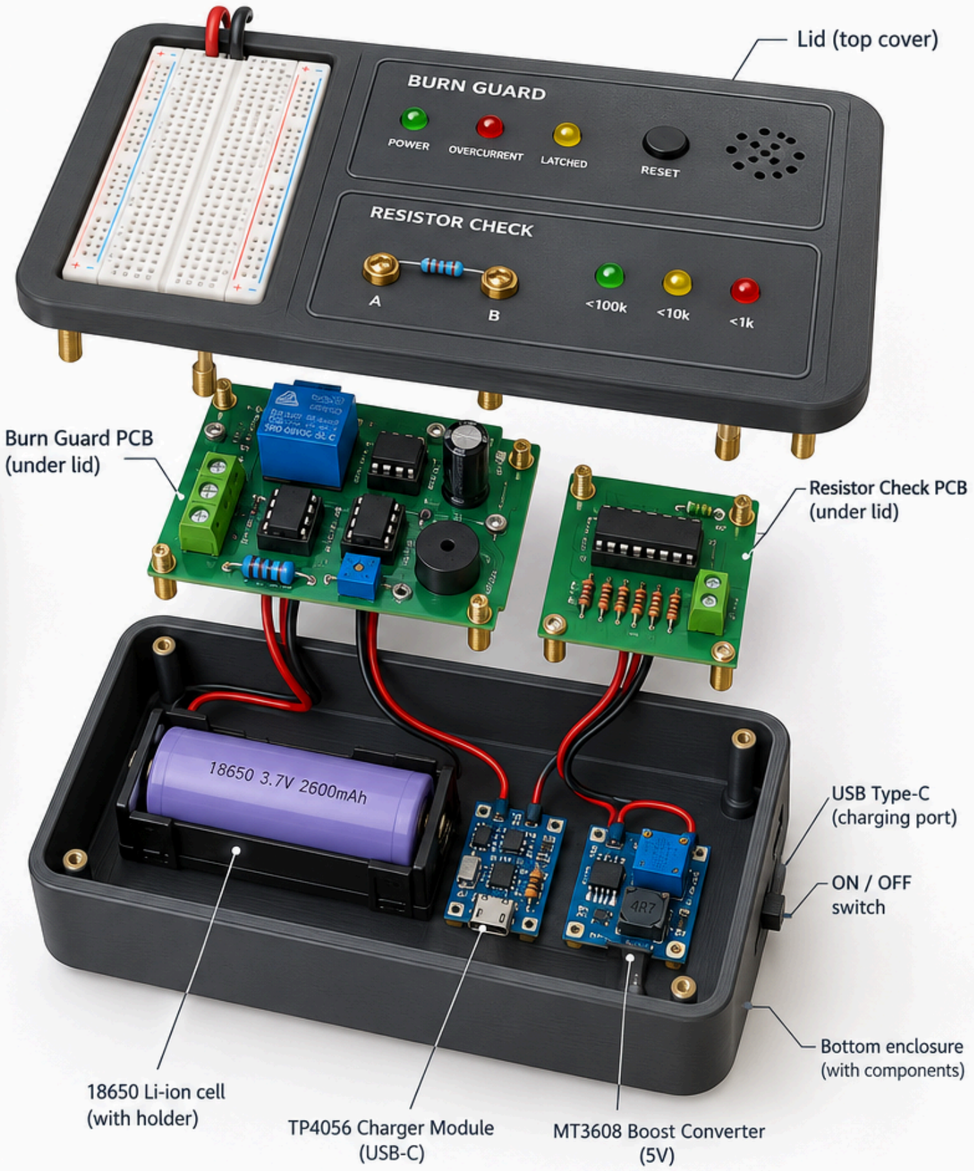


Expected Vision

Lab Bench Debug Kit

Break · Check · Fix





Dimensions (approx.):

Length : 20 cm

Width : 12 cm

Height : 5 cm

Main Features:

- Burn Guard (overcurrent protection)
- Resistor Check (quick value check)
- 18650 battery powered (rechargeable)
- USB Type-C charging
- ON/OFF switch

Key Components:

- 2 × Single-sided PCBs
- 1 × 18650 Li-ion cell
- 1 × TP4056 charger module
- 1 × MT3608 boost converter
- 1 × 5 V relay, ICs, LED indicators
- Brass standoffs & hardware